

Ibn Tofail University

Rigid body mechanics

Problem Set VI

Exercise 1:

1. What is the force necessary to move, at constant velocity, a perfect rigid body, of mass m , having the shape of a cube (Fig 1)?
2. What is the force necessary to move, at constant velocity and without slipping, a perfect rigid body having the shape of a cylinder of radius a and of the same mass m (fig 2)?

We denote by $\vec{R}$ and $\vec{M}_I$ the reduction elements of the torsor of contact actions at point I .

3. Compare the norms of the two forces $\vec{F}_1$ and $\vec{F}_2$.

Correction

Exercise 2:

A perfect and homogeneous solid S having the shape of a disk of mass m and radius a , is in motion, in a vertical plane, on the axis OX of a terrestrial reference frame assumed to be Galilean. Initially, the angular velocity is ω_0 and the velocity of the center of mass G is v_0 .

1. Using the general theorems, determine the differential equations of motion, in x and θ , of S as a function of m , a , and T , where T is the norm of the tangential component of the reaction at point I .
2. Determine the sliding velocity of S with respect to OX and express T as a function of v_g and m .
3. Deduce that the motion ends with a rolling without slipping phase.

Correction

Exercise 3:

A homogeneous hoop S , of mass m and radius R , rolls with slipping on the axis OX , inclined at an angle α with respect to the horizontal, of a terrestrial frame assumed to be Galilean.

1. Calculate the mechanical energy of solid S .
2. By applying the theorem of variation of mechanical energy, find the differential equation of motion of S .

Correction

Exercise 4:

We consider again the system from exercise II of series 5. The revolute joints at points O and C are assumed to be perfect. Initially $\theta = \theta_0$ and $\dot{\theta} = 0$.

1. Determine the kinetic energy of the system with respect to T_1 .
2. Determine the potential energy of the system with respect to T_1 . Deduce the mechanical energy of the system.
3. By applying the angular momentum theorem to disk S_2 at point C , show that the rotation velocity of S_2 is constant.
4. Using the theorem of variation of mechanical energy of the system, find the differential equation of motion, then deduce the period of small oscillations.
5. Determine the period of small oscillations T' in the case where the two solids are rigidly connected. Compare the two periods T and T' .

Correction

Exercise 5:

Consider the system from exercise V of series 5 (hoop and a material point).

1. Find the potential energy of the system.
2. Calculate the power of external actions in two different ways.
3. By applying the kinetic energy theorem to the system, find the differential equation of motion.

Correction